

Phase 1: High-Level Model Proposal

Task Set Information

Goals:

In(apple, drawer);
On(apple, table);...

Objects: Cake, Apple,
Pen, Cabinet, ...

Initial States:

IsHandEmpty(),
IsClose(Objects), ...

Model Proposal

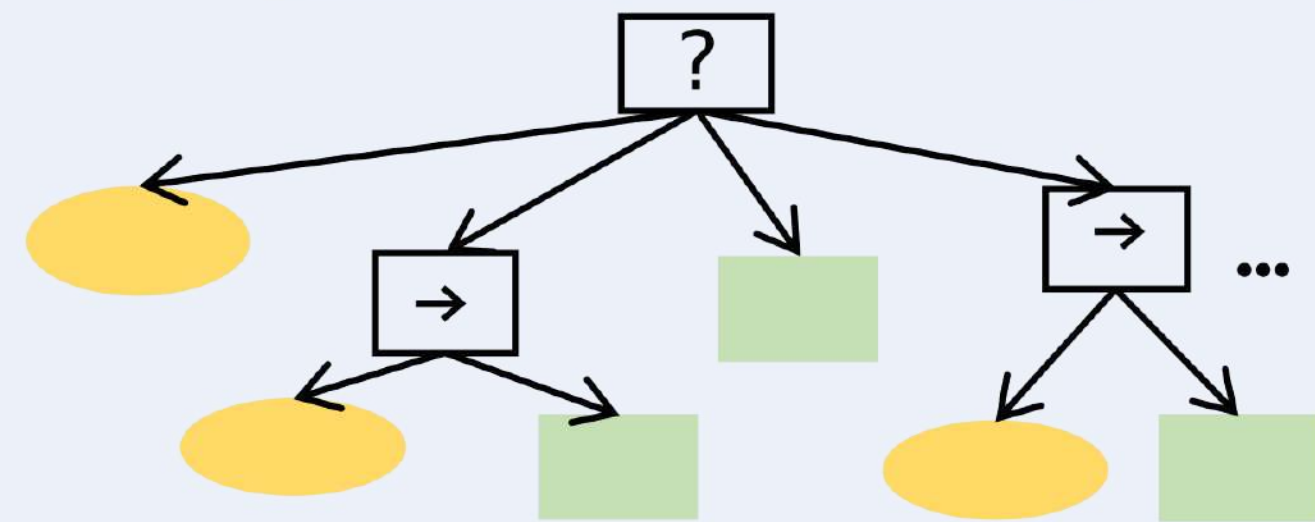


Planning Contexts

Action Models

$h_1 = \langle pre_1, add_1, del_1 \rangle$
 $h_2 = \langle pre_2, add_2, del_2 \rangle$
...

Planning



Phase 2: Low-Level Policy Sampling

Policy Information

Action Model:

$h_i = \langle pre_i, add_i, del_i \rangle$

Policy Space:

- (1) End-to-end policies;
- (2) Hierarchical policies;
- (3) Rule-based policies.

Hyperparameters: args
and prompts for control.

Policy Sampling



Execution Contexts

Sampled policy

Hierarchical policy:
Molmo + Rekep
args: collision buffer 2mm

Policy Execution



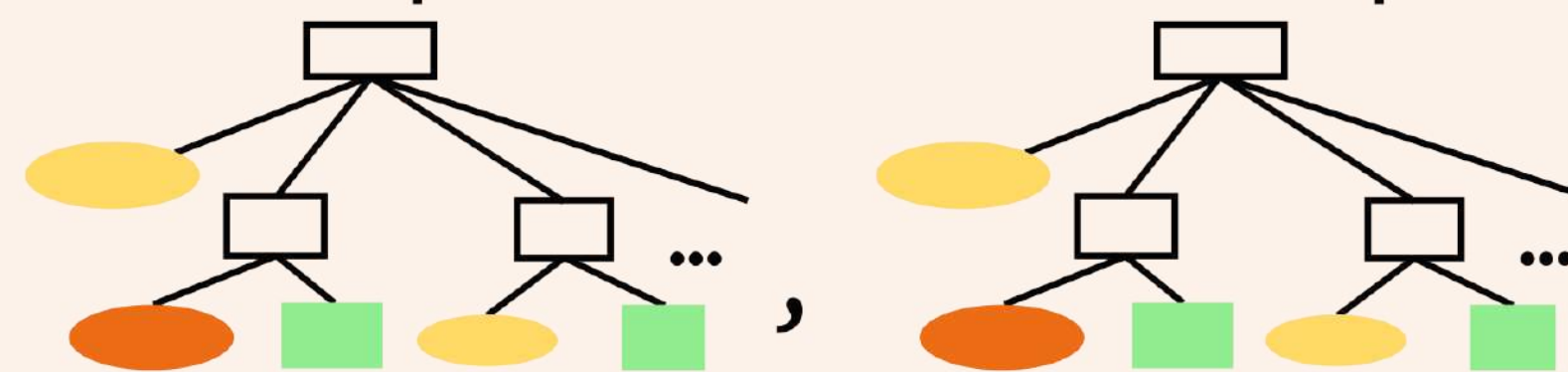
Phase 3: Cross-Level Refinement

h : PlaceIn(apple, drawer)

pre={IsHolding(apple)}
add={In(apple, drawer)}
del={IsHolding(apple)}

Planning Contexts:

This action model is
indispensable in two plans:



Execution Contexts:

Molmo prompts: Point
out the spot inside the
drawer to place an apple.
Camera view: ...

Refinement Prompts

The policy keeps
failing. Check the
contexts and refine
the action model ...

New Action Models:

h'_1 : Open(drawer)

pre={IsClose(drawer)}
add={IsOpen(drawer)}
del={IsClose(drawer)}



h'_2 : PlaceIn(apple, drawer)

pre={IsOpen(drawer)}
IsHolding(apple)}
add={In(apple, drawer)}
del={IsHolding(apple)}



It appears that the
drawer is closed...



Inconsistent X